α-$BaCu_2Se_2$ Thin-Film XRD Analysis and Bayesian Optimization

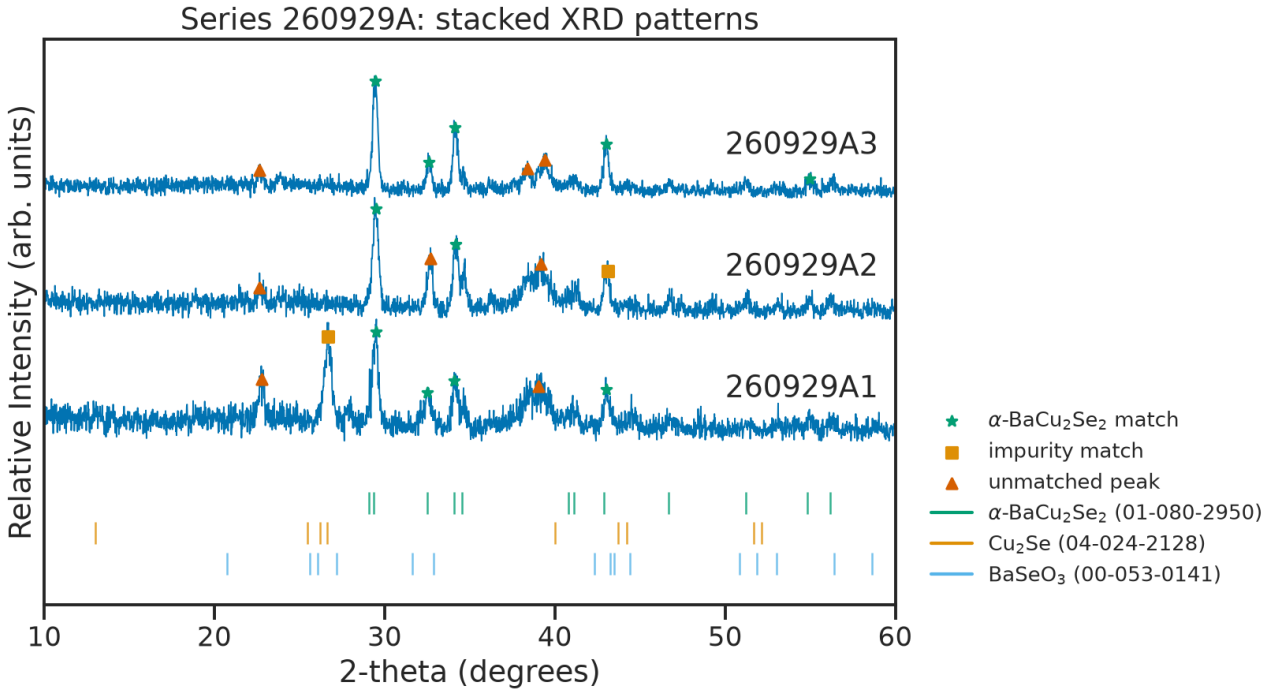
NOMAD-driven data extraction, phase matching, and GP-based synthesis planning

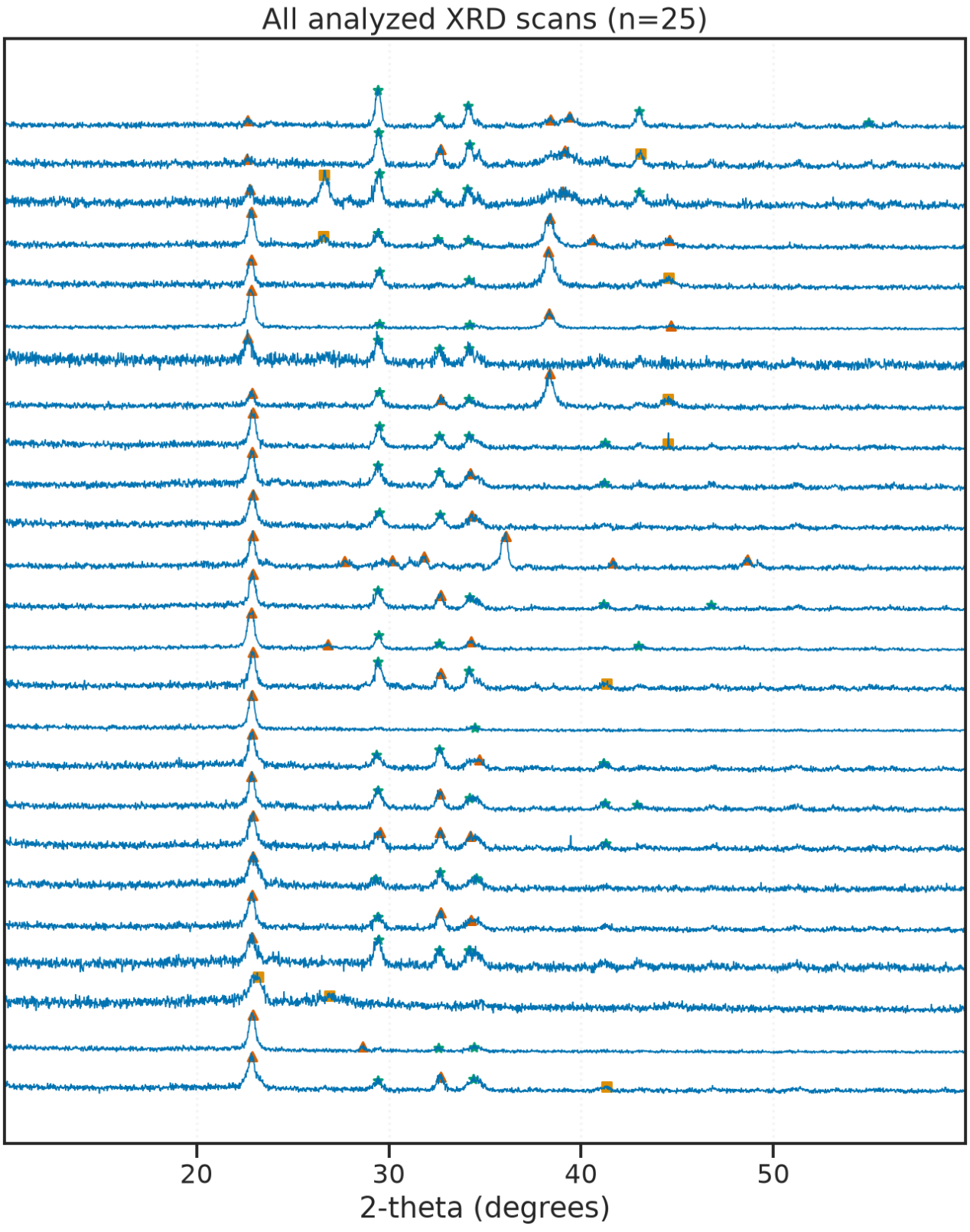
1. What was done

Synthesis records and XRD scans were pulled directly from the NOMAD Oasis REST API for the 42 α-$BaCu_2Se_2$ thin-film samples (INLThinFilmStack entries). Synthesis parameters (spin coating, soft/hard bake, selenization) and selenium-pulse settings were extracted for all samples; 35 include pulse on/off data. For the 25 films with XRD scans, diffraction patterns were matched against ICDD reference cards for the α-$BaCu_2Se_2$ target (01-080-2950) and the Cu_2Se (04-024-2128) and $BaSeO_3$ (00-053-0141) impurities. Phase purity and crystallinity (1/FWHM of the strongest target match) were computed per film. A Gaussian-process Bayesian optimizer was trained on the 21 complete films to recommend the next synthesis conditions, optimizing objective = 0.7 × phase purity + 0.3 × normalized crystallinity over a 7-parameter space (spin speed, bake temperatures, selenization temperature/time, and Se pulse on/off).

2. XRD phase identification

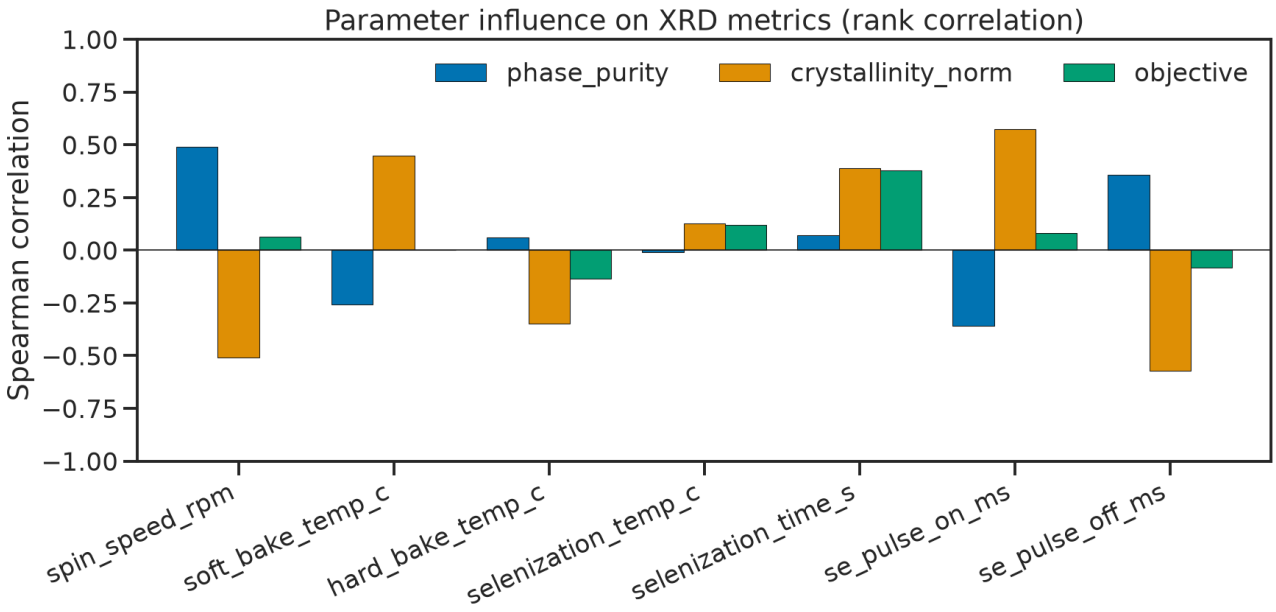
Stacked XRD patterns per series with reference bars for the target and impurity phases. Markers flag target (star), impurity (square), and unmatched (triangle) peaks.

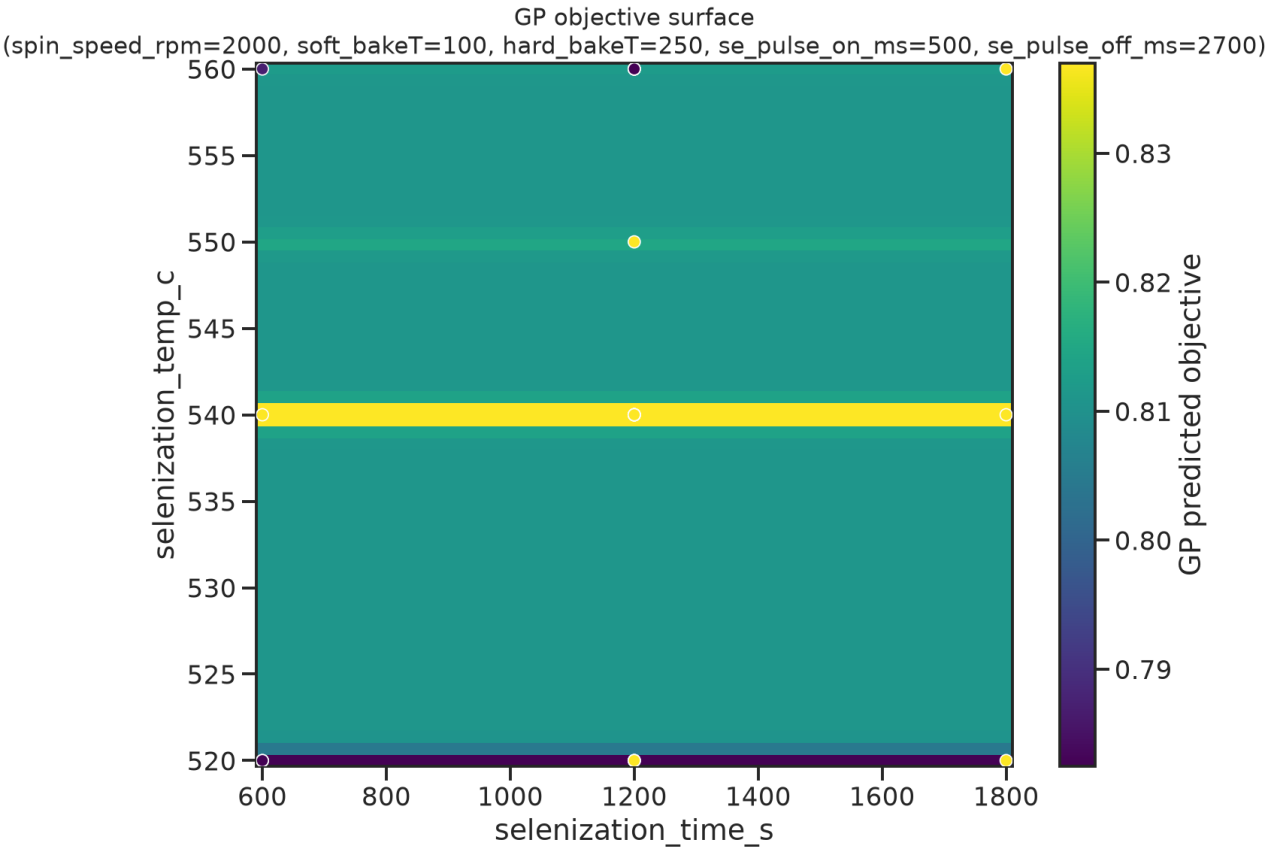




3. Main results

Across the analyzed films the mean phase purity is 0.91 and mean crystallinity is 2.18. The best film is 260429B2 (objective 1.00; purity 1.00, crystallinity 2.92). Rank (Spearman) correlation with the objective identifies selenization time as the dominant synthesis lever; bake temperatures and pulse timing are weaker.



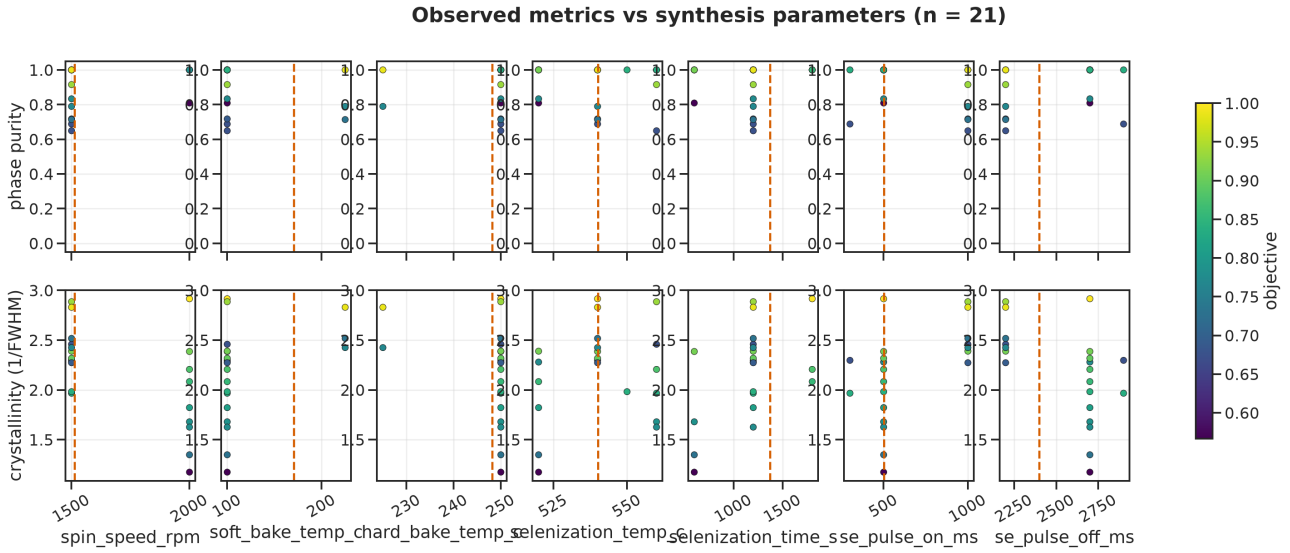


4. Suggested next experiments

Top Gaussian-process suggestions (predicted objective ± std), ranked by expected improvement:

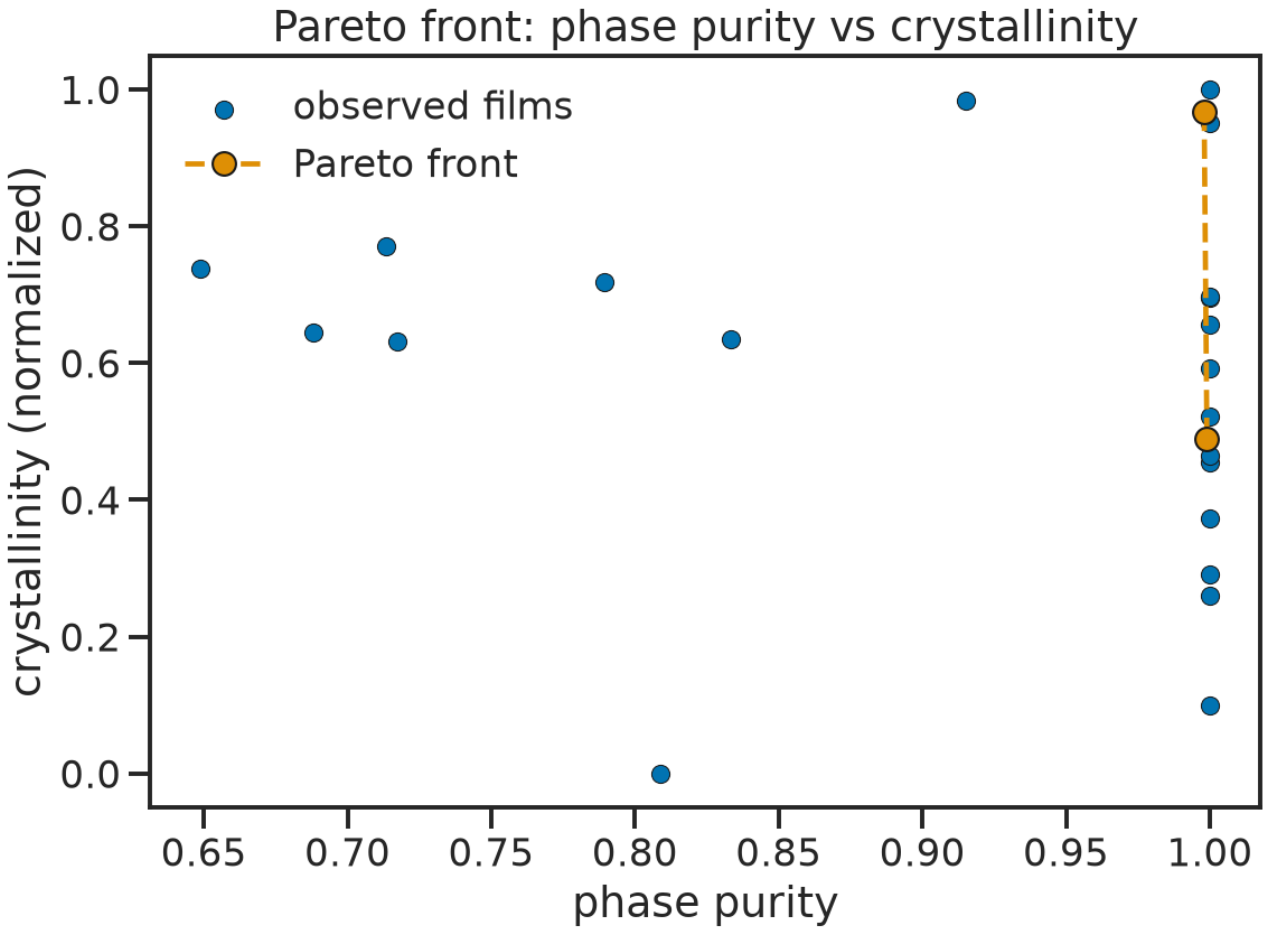
rank	spin (s)peed rpm	soft bake temp (°C)	hard bake temp (°C)	selenization temp (°C)	selenization time (s)	se pulse on (ms)	se pu
1	1512	170	248	540	1372	501	
2	1847	186	245	540	1800	504	
3	1903	221	245	540	1800	519	
4	1603	153	243	538	1800	499	
5	1898	110	239	540	1800	510	

Recommended first experiment: spin 1512 rpm, soft bake 170 °C, hard bake 248 °C, selenization 540 °C for 1372 s with Se pulse 501 ms on / 2398 ms off (predicted objective 0.83).



5. Multi-objective optimization (Ax)

A complementary multi-objective search treats phase purity and normalized crystallinity as two separate objectives instead of one weighted score. Ax (BoTorch) fits a Gaussian process per metric and finds the Pareto front via qNEHVI hypervolume maximization. The suggested conditions below are Pareto-optimal candidates spanning the purity–crystallinity trade-off.



rank	spin (s)peed rpm	soft bake temp (°C)	hard bake temp (°C)	selenization temp (°C)	selenization time (s)	se pulse on (ms)	se pu
1	1916	183	247	539	1791	601	
2	2000	100	250	544	1800	659	
3	2000	100	250	537	1800	399	
4	1842	100	238	538	1800	596	
5	2000	197	238	545	1800	434	